Name: Charné Stasie
Student Number: 2805001596
Data Analytics & Data Science
Research Assignment 1

1. A database is a well-organized set of data that is stored in a way that makes it easy to get to, manage, and change. It lets users or systems save information in an organized way so that they can get it back quickly when they need it.

Databases are important because they keep a lot of information safe and let businesses use that information to make choices. It would be slow and unreliable to manage information without databases.

For example, banks use databases to keep track of their customer's accounts, balances and transactions. Example 2: Websites that let you shop online keep track of ~~the~~ products, customer information and past orders.

2. A data warehouse is a system that stores a lot of historical data from different sources in one place so that it can be analyzed and reported on.

A regular database is used for everyday tasks like transactions, while a data warehouse is used to look at data over time.

Companies look at trends, patterns, and performance

in data warehouses to help them make better business decisions.

3. A data lake is a place to store raw data in its original form, whether it is structured or not.

A data warehouse stores clean and organized data, but a data lake stores raw and unprocessed data.

When a business needs to store a lot of different kinds of data for later use, especially for advanced analytics, it would use a data lake.

4. A data mart is a smaller part of a data warehouse that is only for one department like sales or finance.

It takes data from a data warehouse, but it is made for a certain group of people.

It is most often used by departments or teams that need quick access to important information.

5. Tables are used to organize and store structured data. for example, Excel spreadsheets.

Semi-structured data has some structure, but it doesn't have a strict format. for example, Json files.

There is no clear format for unstructured data. For example, pictures or videos!

6. Data modeling is the process of figuring out how to store and organize data in a system.

It is important because it makes sure the system works well, is accurate, and is simple to use.

There are two kinds of data models:
A conceptual model shows a big picture of data.
A logical model shows the structure in detail, such as relationships and fields.

7. Data mining is the act of looking at large amounts of data to find patterns and useful information.

Companies use it to make better choices and learn more about how customers act.

Example 1: Retail stores use data mining to suggest things to buy.
Example 2: Banks use it to find out if someone is lying.

8. A DBMS is a piece of software that lets people make, manage and use databases

People use MySQL for web apps because it is easy to use.
PostgreSQL is strong and can handle complex queries.
Big companies use Oracle for complicated systems.
MongoDB is a good database for modern apps because it can store data in different ways.

9. ETL stands for "Extract, Transform Load."

Collecting data from different sources is what extract means. Transform means getting the data ready and cleaning it.
Load means putting data into a system, like a data-warehouse.

This step is important because it makes sure that the data is clean and ready to be analyzed.

10. CSV is a simple way to store data in rows and columns. It is used with small amounts of data.
JSON is a way to store and share data between different systems.
Parquet is good for big data because it is quick and works well.
Big data systems use Avro to store and move data.

11. Artificial Intelligence is when machines can do things that normally need human intelligence.

Narrow AI is made for certain jobs, like voice assistants. Siri is an example.
General AI is a type of AI that can do any mental task like a person, but it isn't fully developed yet.

12. Machine learning is a kind of AI that lets systems learn from data instead of being programmed step to step.

Machine learning gets better with more data, while traditional programming uses fixed rules.

Supervised learning needs data that has been labeled.
Unsupervised learning finds patterns in data without any help.
Reinforcement learning uses rewards and punishments to teach.

13. Deep learning is a type of machine learning that uses neural networks to learn from a lot of data.

Neural networks are computers systems that work like the brain and process data in layers.

They work by sending data through layers and changing things based on the results to make them more accurate.

14. Natural Language Processing is a branch of AI that helps computers understand and respond to what people say.

For example:
Bots
Voice helpers
Apps for translating.

15. A Large Language Model is an AI model that learns to understand and create language by reading a lot of text.

It can handle more difficult language tasks than regular models.

ChatGPT and Claude are two examples. You can use them to write, answer questions, and get help with coding.

16. Generative AI makes new things like text, pictures and music.

It learns from data and makes content that is similar to what it has seen before.

For example:
ChatGPT makes text
DALL E makes pictures.
Midjourney makes art.

17 The data used to teach an AI model is called a training dataset.

Quality is important because bad data can give you bad results.

The model can give unfair or wrong results if the data is biased or wrong.

18. The ability of machines to comprehend pictures and movies is known as computer vision.

Algorithms are used by machines to identify items

and patterns in pictures.

First example: facial recognition.
Example 2: Autonomous vehicles that recognize items.

19. An API is a tool that facilitates communication between various software systems.

A waiter in a restaurant who takes your order to the kitchen and returns your meal is a basic example.

Because they enable systems to collaborate and exchange data, APIs are crucial.

20. A code used to identify and manage access to an API is called an API key.

It must be kept private and kept in a secure location.

If the API is made public, it could be abused by others, resulting in data loss or additional expenses.

21. REST stands for Representational State Transfer.

GET retrieves data
POST sends new data
PUT updates data
DELETE removes data

22. JSON is a format used to store and exchange data. It is simple to write and read.

Example:
"age": 25; "name": "John"

23. The ability to write precise instructions for AI tools is known as prompt engineering.

Bad prompt: Describe your business.
An improved prompt: Describe how data is used by small firms to increase sales.

Bad prompt: Write a piece about artificial intelligence.
An improved prompt: Write a brief paragraph outlining AI in clear language for novices.

24. Every item in a list is given a function by a map function.

For instance, in Python:
numbers = [1, 2, 3] result = list (map(lambda x: x * 2, numbers))
The result will be [2, 4, 6].

25. Cloud computing is the delivery of computing services over the internet.

IaaS offers servers and other infrastructure. For instance, AWS EC2 PaaS offers development platforms. For instance, Google App Engine SaaS offers applications via the internet. For instance, Google Drive.

26. Code modifications are monitored by a version control system.

Git is popular because it facilitates change tracking and collaboration.

A saved change is called a commit.
A branch is a distinct version of the project used for development or testing.

27. A Jupyter Notebook is a tool used to write and run code with notes and visuals.

It is mostly used with Python.

It is well-liked because it makes data testing, visualization, and explanation simple.

28. Python is a programming language renowned for its simplicity and readability.

It is used in data analytics since it has numerous essential libraries.

Pandas is utilized for data analysis.
NumPy is used to perform numerical operations.
Matplotlib is used for charting
Scikit Learn is used for machine learning.

29. Data privacy refers to the safeguarding of personal information from misuse.

It is critical because users must ~~trus~~ trust how their data is used.

GDPR is a European Union law that protects personal data and applies to enterprises that handle data for EU individuals.

30. The concept of data ethics refers to the responsible and fair use of data.

Example 1: Using personal information without authorization.
Example 2: Biased hiring systems.
Example 3: Selling user data without their consent.

These are troublesome because they violate both privacy and fairness.

31. Data governance is the process of managing data to ensure its accuracy, security and appropriate use.

Data quality, security, policies and compliance are all important considerations.

32. Bias in AI refers to the model producing unfair findings owing to skewed data.

Consider a recruiting system that benefits specific groups.

This can result in discrimination and biased decisions.

33.

Role	Focus	Skills	Tools
Data Analyst	Analyze data	Excel, SQL	Power BI
Data Scientist	Build models	Python, ML	Python, R
Data Engineer	Build systems	Databases	Spark, SQL
AI Engineer	Build AI systems	AI, ML	Tensorflow

34. Business intelligence is the utilization of data to help make business choices.

Power BI and Tableau are tools for creating reports and dashboards that help businesses analyze data.

35. Machine learning was the most exciting topic to me since it demonstrated how systems can learn from data and improve over time without being designed step by step. It helped me realize how powerful data is.

36. The most difficult topic was deep learning, which incorporates complicated systems such as neural networks. I used online tutorials and videos to better understand how they work.

37. Data and AI will influence my future by enhancing the way I work and make decisions. For example, I may examine trends with data tools and automate jobs with AI technologies, making my work faster and more productive.

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